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Object-Oriented Programming (University of KwaZulu-Natal)



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COMP200: Practical 1

Battleships is a game played between the computer and a player on a 10×10 grids of squares. The grid represents the ocean on which a fleet of ships is positioned. The ships have different lengths. The fleet of ships consists of an aircraft carrier (five squares long), a battleship (four squares), a destroyer (three squares), a submarine (two squares), and a PT boat (one square). The computer places the ships randomly on the grid. In this version of the game, the ships are placed horizontally. The player cannot see where the ships are placed. The objective of the game is for the player to sink all the ships by firing on them. The player fires on one square at time. If the square contains part of a ship then this is a hit, otherwise it is a miss. After firing on the square, the computer marks the square with:

- a 'H' (Hit: the square contains part of a ship)
- a 'M' (Miss: the square does not contain part of a ship)

A sample run of the game is shown below:

```
    Welcome to BattleShips:
      0  1  2  3  4  5  6  7  8  9
0     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
1     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
2     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
3     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
4     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
5     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
6     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
7     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
8     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
9     [ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9): 5
    col (0-9): 6

      0  1  2  3  4  5  6  7  8  9
0     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
1     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
2     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
3     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
4     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
5     [ ][ ][ ][ ][ ][M][ ][ ][ ]
6     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
7     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
8     [ ][ ][ ][ ][ ][ ][ ][ ][ ]
9     [ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9): 3
    col (0-9): 2
```


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```
      0  1  2  3  4  5  6  7  8  9
0    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
1    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
2    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
3    [ ][ ][H][ ][ ][ ][ ][ ][ ][ ]
4    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
5    [ ][ ][ ][ ][ ][M][ ][ ][ ][ ]
6    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
7    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
8    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
9    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9): 2
    col (0-9): 3


      0  1  2  3  4  5  6  7  8  9
0    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
1    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
2    [ ][ ][M][ ][ ][ ][ ][ ][ ][ ]
3    [ ][ ][H][ ][ ][ ][ ][ ][ ][ ]
4    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
5    [ ][ ][ ][ ][ ][M][ ][ ][ ][ ]
6    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
7    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
8    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
9    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9): 3
    col (0-9): 3


      0  1  2  3  4  5  6  7  8  9
0    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
1    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
2    [ ][ ][M][ ][ ][ ][ ][ ][ ][ ]
3    [ ][ ][H][H][ ][ ][ ][ ][ ][ ]
4    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
5    [ ][ ][ ][ ][ ][M][ ][ ][ ][ ]
6    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
7    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
8    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
9    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9): 3
    col (0-9): 4
```


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```
      0  1  2  3  4  5  6  7  8  9
0    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
1    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
2    [ ][ ][M][ ][ ][ ][ ][ ][ ][ ]
3    [ ][ ][H][H][H][ ][ ][ ][ ][ ]
4    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
5    [ ][ ][ ][ ][ ][M][ ][ ][ ][ ]
6    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
7    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
8    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]
9    [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

    What row and column do you want to fire on?
    row (0-9):
    col (0-9):

// continued ...
```

The following is a possible design. You should have at least two classes: `BattleShipsApp` and `BattleShips`. The `BattleShipsApp` contains a main method that runs the game while the other class provides the functionality.

The `BattleShips` class has an attribute: a 2-dimensional array `grid` of type **char** maintains the state of the grid as the game is played. Before the game starts the **constructor** places the various ships randomly on the grid. When the player fires on a square the method `processFire` determines if there is a hit or miss.

The following methods should be written in the class `BattleShips`:

1. `print()` that prints out the current state of the grade. It prints out a blank square if there is no hit or miss in that array position, otherwise it prints the character in that position.

2. `placeShips()` that randomly places the ships on the grid. It does this by assigning a 'S' character in each slot of the array that contains an item of the ship. The method first picks two random numbers to represent the start position of the ship and then fills the appropriate number of slots in the array with 'S' characters.

3. `canFire(int row, int col)` that returns true if the square specified by `row` and `col` has not been fired on before, false otherwise.

4. `processFire(int row, int col)` that is given the row and column of the square in which the player wants to fire. The method then assigns a 'H' to this square if it contains part of a ship or it assigns a 'M' if it does not.

5. `gameOver()` that returns true if all ships have been hit, false otherwise.

   Think carefully about the method signatures for each method

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